

AI-Assisted Cervical Cancer Cytology Analysis Report

Schema Version: 2.1.0

AI Summary

Report ID	N/A
Slide ID	raw
Analysis Timestamp	2026-04-10 12:45:22
AI Screening Result	AI Screening Result: High Risk (Requires Expert Review)
Total Cells Analyzed	1517
Abnormal Cells	530
Abnormal Cell Ratio	34.9%
Recommendation	Urgent expert review by a cytotechnologist/pathologist is recommended.

Specimen Adequacy

Cellularity: **ADEQUATE**. Specimen is marked ADEQUATE because approximately 1517 analyzable epithelial-cell candidates were detected, which is generally sufficient for AI screening interpretation.

Quantitative Analysis

Cell Classification	Count	Clinical Group
Koilocytotic	523	ABNORMAL
Low Confidence Predictions	386	LOW_CONFIDENCE
Superficial Intermediate	230	BENIGN
Parabasal	214	BENIGN
Metaplastic	157	BENIGN
Dyskeratotic	7	ABNORMAL

Low Confidence Predictions: 386 cells were not confidently classified and are shown as uncertain candidates. These entries should be interpreted with extra caution.

Detected Abnormal Cells

The table below lists top AI-selected abnormal-morphology candidates. Confidence is a model score and does not represent diagnostic certainty.

Cell ID	Class	Confidence	Coordinates (BBox)
13	Koilocytotic	98.92%	[1242, 677, 1437, 869]

10	Koilocytotic	98.86%	[1303, 225, 1629, 584]
48	Koilocytotic	98.59%	[0, 552, 142, 714]
30	Koilocytotic	98.56%	[836, 325, 1080, 618]
65	Koilocytotic	98.54%	[1057, 798, 1200, 979]
94	Koilocytotic	98.49%	[455, 1278, 626, 1468]
52	Koilocytotic	98.41%	[935, 861, 1082, 1042]
73	Koilocytotic	98.39%	[450, 924, 583, 1062]

Visual evidence files are not available in this run. Bounding-box coordinates are provided for traceability.

Clinical Interpretation (AI-Assisted)

AI-assisted interpretation: abnormal-appearing cells are estimated at 34.9% (530/1517) of analyzed cells. Koilocytotic candidates: 523. Koilocytotic morphology may reflect HPV-related cytopathic change and can be suggestive of LSIL in the proper clinical context. Dyskeratotic candidates: 7. Dyskeratotic morphology may be suspicious for higher-grade squamous abnormality and can be suggestive of HSIL risk. These findings are AI suggestions and must be reviewed and confirmed by a qualified cytotechnologist or pathologist.

Primary AI Finding: High burden of abnormal cells (34.9%).

AI Model Information

Model	MobileViT-S (CNN-Transformer Hybrid)
Model Type	MobileViT-S
Model Version	mobilevit_s_sipakmed_stain_normalized
Training Data	SIPaKMeD
Accuracy	92.60%
Precision	93.00%
Recall	93.00%
F1-Score	93.00%
Confidence Threshold	75.00%

Limitations

- Model performance is bounded by training data composition and may not generalize to all laboratory protocols.
- False positives and false negatives are possible; model confidence is not equivalent to diagnostic certainty.
- Staining variability, preparation artifacts, and overlap/debris can affect detection and classification quality.
- Low-cellularity or poor-quality samples may reduce reliability and should be interpreted with caution.

Disclaimer (Mandatory)

This report is generated by an AI-based screening system and is intended for research and preliminary analysis only. It is not a diagnostic tool. All findings must be reviewed and confirmed by a qualified cytotechnologist or pathologist before any clinical decision is made.